MILITARY STANDARD

MIL-STD-1386-47
(NAVY)

24 FEBRUARY 1975

LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

MINE, UNDERWATER MK 52 AND MODS
IN CRATE MK 52
FLEET ISSUE UNIT LOAD

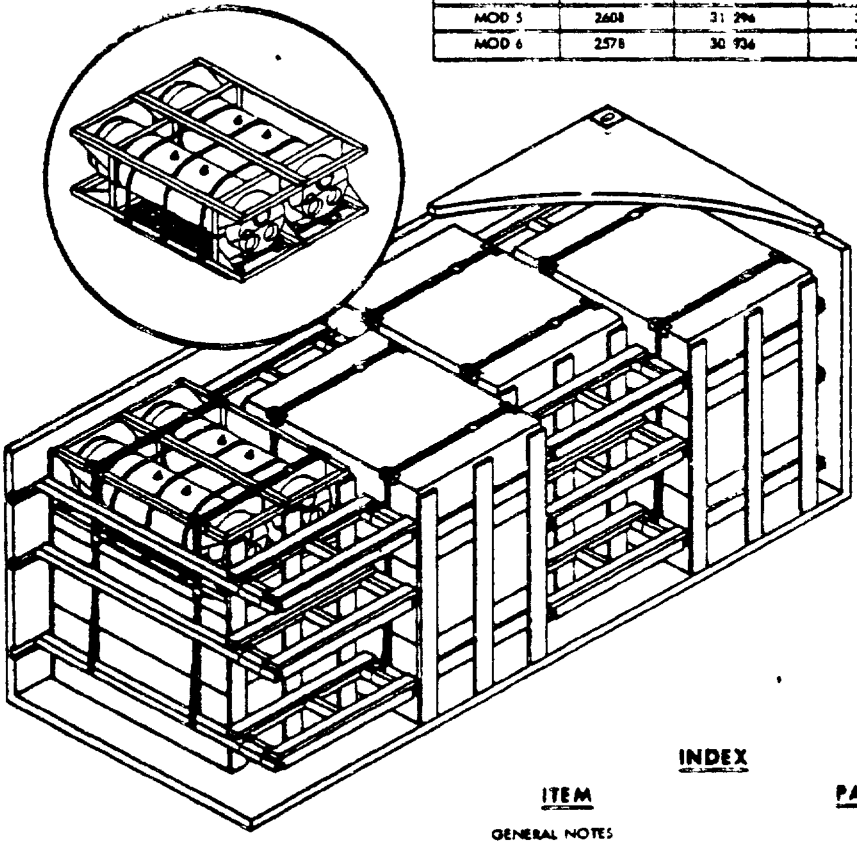
UNIT LOAD DATA

UNIT LOAD DRAWING NR-64/208
DIMENSIONS .... 66 1/4 L x 39 3/4 W x 27 3/4 H
CUBE .... 53.9 CU FT
WEIGHT .... SEE TABLE I

HAZARD CLASSIFICATION:
DOT - EXPLOSIVES A
CG - CLASS X-A

TABLE I

MK 52 MINE UNDERWATER	UNIT LOAD WT (LBS)	CONTAINER LOAD WEIGHT (LBS)	CONTAINER GROSS WEIGHT (LBS)
MOD 1	2 388	29, 336	35, 114
MOD 2	2588	31, 056	37, 314
MOD 3	2612	31, 344	37, 602
MOD 4	2524	30, 288	36, 546
MOD 5	2608	31 296	37, 354
MOD 6	2578	30 936	37, 194



INDEX

ITEM	PAGE
GENERAL NOTES	2
COFC, TOFC CONTAINER-SHIP & HIGHWAY	2 & 3

NOTES:

1 UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.

2 FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CARLOADING MILITARY STANDARDS, REFER TO THE INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED FOR TRAILER-ON-FLAT-CAR (TOFC), CONTAINER-ON-FLAT-CAR (COFC), CONTAINERSHIP AND HIGHWAY SHIPMENTS

M.S. Gray NAVSEA 2/10/75
SIGNATURE TECHNICAL DIRECTION [illegible] (TDA) DATE

2/20/75
SIGNATURE NAVSEASYSCOM BY DIRECTION DATE

APPROVED BY US COAST GUARD

CAPT USCG
11-7-74

APPROVED BY BUREAU OF EXPLOSIVES

11/27 4

ORIGINATOR 2/21/75
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE, NEW JERSEY

GENERAL NOTES

1 THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH THE UNDERWATER MINE MK 52 AND MODS IN CRATE MK 52 WHEN THE MINES ARE PALLETIZED IN ACCORDANCE WITH WR-54/208

2. THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD-BRACING SYSTEM THAT MEETS THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C

3. IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6C THE MILVAN CONTAINER HAS BELT RAILS AT THE 72" AND 83" HEIGHTS FROM THE CONTAINER FLOOR.

4 CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBERS. UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED INTO BELT RAILS FOR SHIPMENT

CAUTION

WHEN POSITIONING CROSSMEMBERS MAKE A VISUAL CHECK TO ENSURE THAT CROSSMEMBERS ARE LOCKED SECURELY IN PLACE.

3 WHEN THE LONGITUDINAL SLACK BETWEEN A CROSSMEMBER AND THE UNIT LOAD IS GREATER THAN ONE INCH, INSTALL SHIM PIECE 9 TO THE CROSSMEMBER TO FILL THE VOID TWIST TIE WITH THREE PIECES OF 16 GAUGE SOFT ANNEALED IRON WIRE, AS SHOWN IN DETAIL A THIS SHIM STRIP MAY BE ADDED OR OMITTED AS REQUIRED.

6. SWAY BRACE, DETAIL D, IS TWIST TIED TO THE CROSSMEMBERS AT BOTH ENDS WITH 16 GAUGE SOFT ANNEALED IRON WIRE. LENGTH OF STRUT, PIECE 4, SHALL BE OF SUFFICIENT THICKNESS TO CAUSE THE SWAY BRACE TO FILL THE VOID BETWEEN THE UNIT LOAD AND THE CONTAINER SIDE WALL

7 EACH BAY OF UNIT LOADS IS TOMMED DOWN WITH TWO 1 1/4 x .035 STEEL STRAPS. STRAPS ARE ATTACHED TO THE CROSSMEMBERS AS SHOWN IN SECTION B-B. SECURE EACH END OF STRAP WITH ONE SEAL WITH TWO PAIR OF NOTCHES PER SEAL OR TWO SEALS, BUTTED TOGETHER WITH TWO PAIR OF CRIMPS PER SEAL

3 APPLICABLE MATERIAL SPECIFICATIONS:
DUNNAGE LUMBER - FED. SPEC. MM-L-751.
NAILS - FED SPEC FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT.
STRAP - FED. SPEC. QQ-S-781, TYPE I, CLASS A.

9 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA, CLOSE AND SEAL CONTAINER DOORS AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 (VOL. 1).

TOFC/COFC,CONTAINERSHIP AND HIGHWAY

1 THE CONTAINER LOAD CONSISTS OF 12 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS DOCUMENT

NOTE

THE TOP LAYER OF THE LAST BAY MAY BE OMITTED IF CONTAINER HEIGHT PREVENTS LOADING. REMAINING MINES IN THE BAY ARE TOMMED DOWN USING A CROSSMEMBER ON THE 60" HEIGHT. SHIM BETWEEN CROSSMEMBER AND MINE CRATE IF REQUIRED

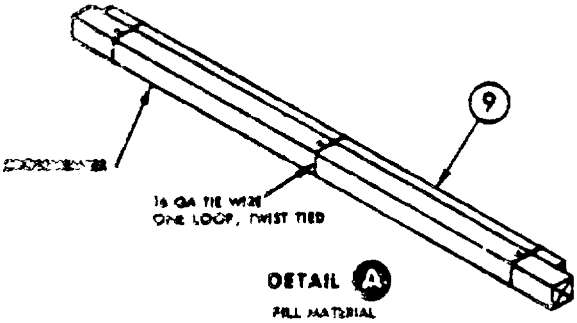
2. THE LOADED MILVAN CONTAINER SHALL BE ON A 20 FT MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES WHEN BEING MOVED IN TOFC SERVICE

3. WHEN THE LOADED CONTAINER IS IN TOFC OR COFC SERVICE, IT SHALL BE ON FLAT RAILCARS THAT COMPLY WITH BUREAU OF EXPLOSIVES PAMPHLET 6C

4. THE LOADED MILVAN MAY BE TRANSPORTED OVER PUBLIC HIGHWAYS ON A MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES OR ON SUITABLE COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY

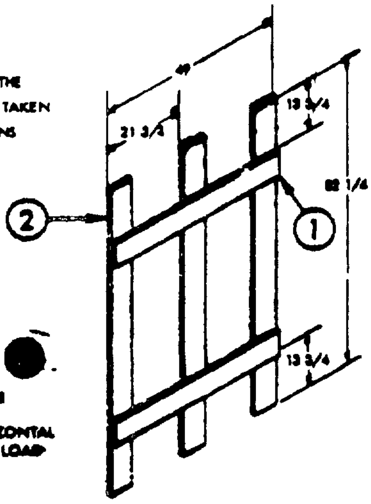
NOTE

IN CERTAIN STATES THE GROSS VEHICLE WEIGHT (G.V.W.) IS LIMITED BY THE DISTANCE BETWEEN THE AXLES OF THE ASSEMBLED RIG. CARE SHOULD BE TAKEN TO SEE THAT THE PROPOSED G.V.W. DOES NOT EXCEED THESE LIMITATIONS



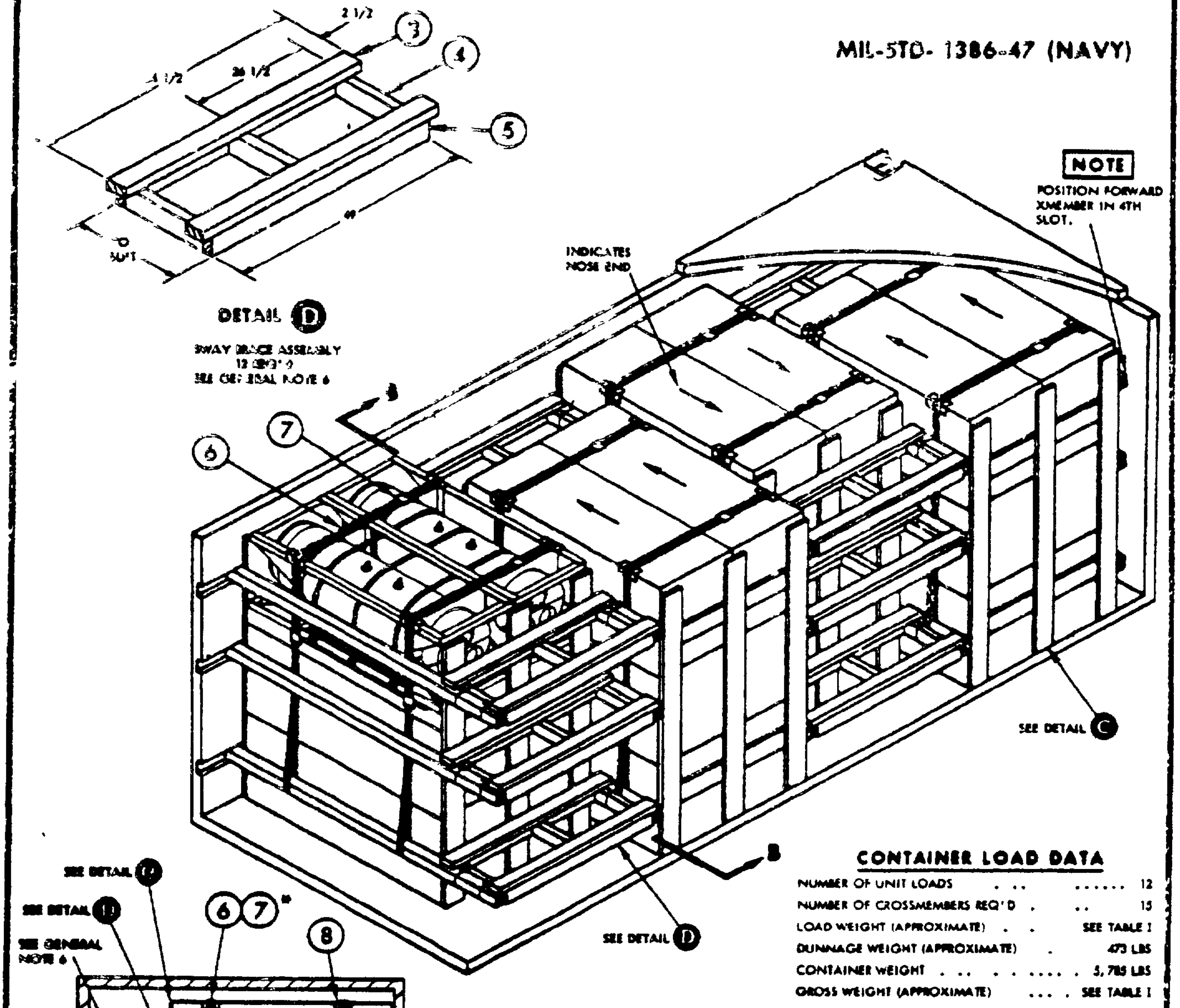
DETAIL A

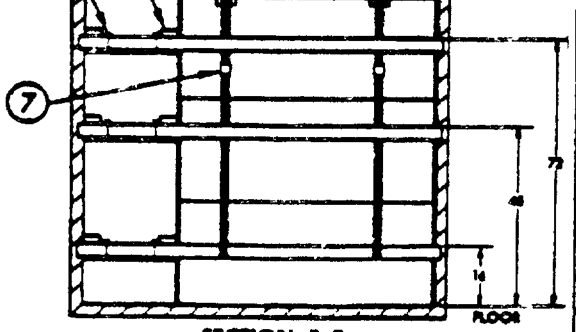
FILL MATERIAL
SEE GENERAL NOTE 5



DETAIL D.

SIDE FRAME
9 REQ'D
POSITION HORIZONTAL
TOWARD UNIT LOAD





SECTION B-B

* FAR END OF STRAP IS ATTACHED TO FORWARD 72 INCH HIGH CROSSMEMBER OF THE BAY. STRAP SEAL IS LOCATED SO THAT IT WILL BE ON TOP OF THE UNIT LOAD WHEN STRAP IS TENSIONED.

CONTAINER LOAD DATA

NUMBER OF UNIT LOADS	12
NUMBER OF CROSSMEMBERS REQ'D	15
LOAD WEIGHT (APPROXIMATE)	SEE TABLE I
DUNNAGE WEIGHT (APPROXIMATE)	473 LBS
CONTAINER WEIGHT	5,785 LBS
GROSS WEIGHT (APPROXIMATE)	SEE TABLE I

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
9	SHIM	4 WIDE x 90 LONG x THICKNESS TO SUIT	AS REQ'D	SEE GENERAL NOTE 5		
8	EDGE PROTECTOR	2 3/4 x 2 x .030	16	-	-	-
7	SEAL	1 1/4	16	-	-	-
6	STRAP	1 1/4 x .035 x 16 FT	8	-	-	-
5	LONGITUDINAL	2 x 4 x 49	24	4	2/JOINT	16d
4	STRUT	2 x 4 x TO SUIT	36	SEE 5	-	-
3	SUPPORT	2 x 4 x 54 1/2	24	5	7	10d
2	VERTICAL	1 x 6 x 82 1/4	24	1	3/JOINT	8d CLINCHED
1	HORIZONTAL	1 x 6 x 49	16	SEE 2	-	-

LIST OF MATERIALS AND NAILING DATA

TOFC/COFC, CONTAINERSHIP AND HIGHWAY

REVIEW ACTIVITY
NAVY - OS

PREPARING ACTIVITY
NAVY - OS
PROJECT NO 8140-N259

SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No 22-R333

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION
MIL-STD-1386-47 (Navy)

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

S/N-0102 014-1801 0-38224

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Naval Weapons Station Earle (NWHL- R. Harris)
Colts Neck, New Jersey 07722

FOLD